3D Design, Kinematics & FEA: 2.0L Inline-4 Automotive Engine

Role: Mechanical Designer / CAD Engineer (Parametric Modeling, Assembly, Structural Analysis)
Tech Stack: Autodesk Fusion 360 (Design, Assembly & Simulation Workspaces), Mechanical Drafting.

CHALLENGE AND OBJECTIVE

The objective of this project was to engineer a complete digital twin of a 2.0L 4-cylinder combustion engine from scratch. The project spanned the entire mechanical design lifecycle: translating 2D technical blueprints into precise 3D parametric models, defining complex kinematic relationships to validate internal movement, and concluding with structural FEA simulations to test component viability under peak combustion loads.

ARCHITECTURE AND SPECIFICATIONS

Engine Configuration:

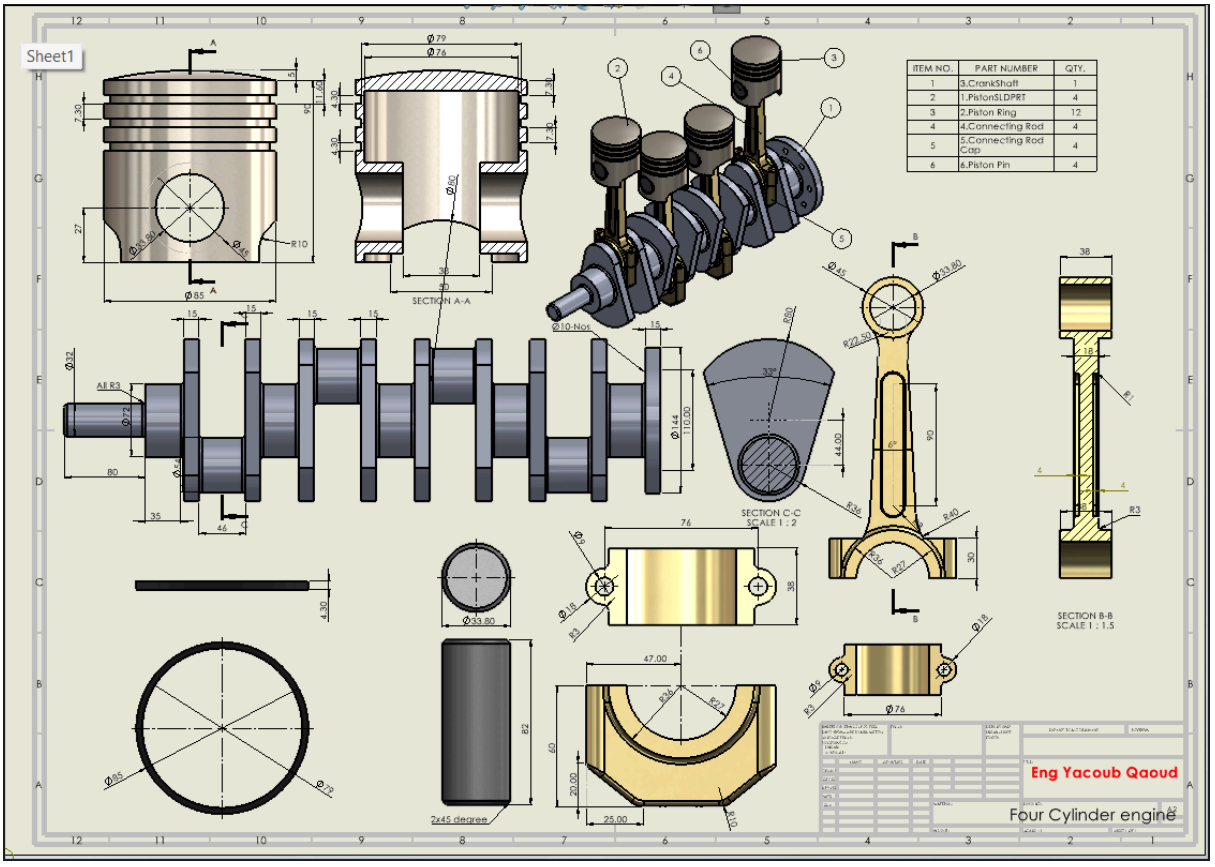
- **Type:** Inline 4-Cylinder (I4), ~2.0 Liters displacement.
- **Dimensions:** 80mm piston diameter, optimized for standard automotive applications.
- **Modeled Components:** Crankshaft, connecting rods, rod caps, piston heads and pins
- **Software & Environment: CAD/CAE**
- **Tool:** Autodesk Fusion 360.
- **Workspaces:** Design (Solid Modeling), Assembly (Joints/Motion), Simulation (Static Stress).

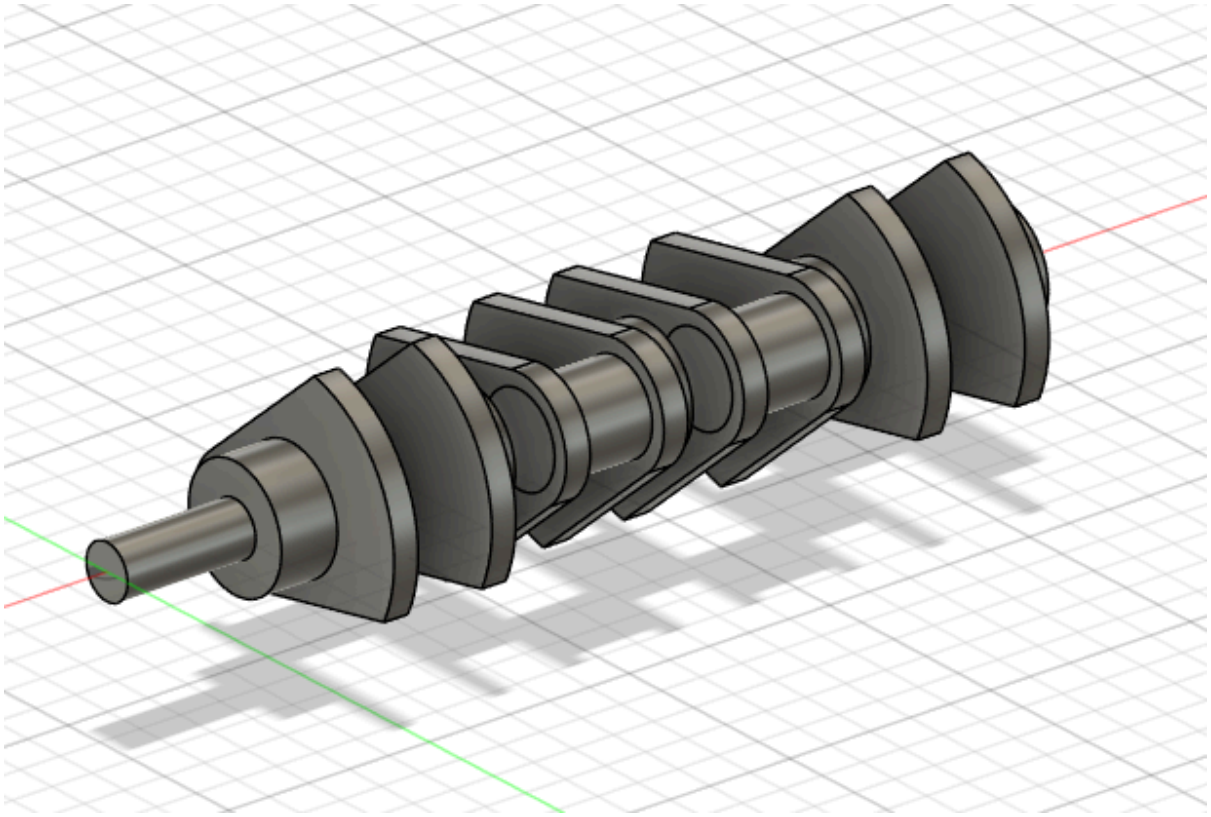
ENGINEERING PROCESS

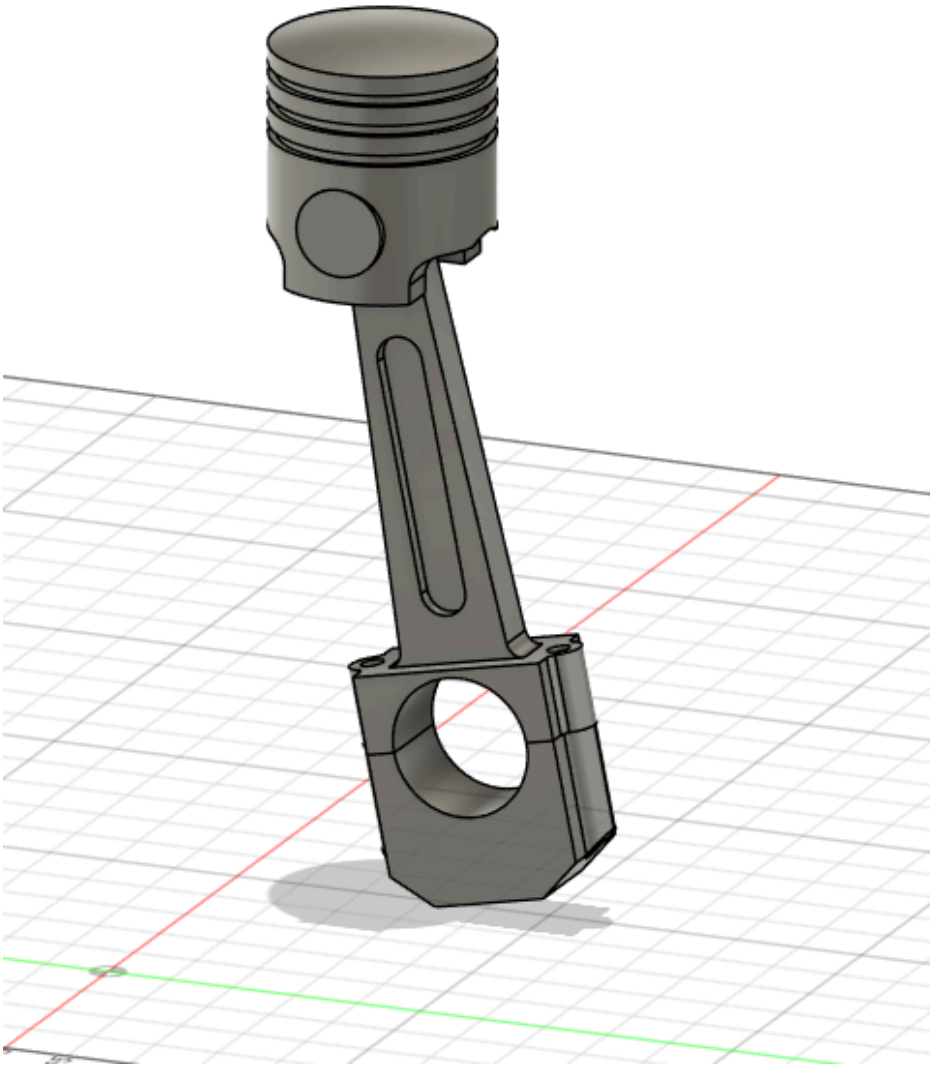
1. Parametric 3D Modeling (CAD)

The foundational step involved modeling each individual component with strict dimensional tolerances based on standard automotive blueprints.

- **Approach:** I utilized sketch constraints, revolving features, and precise extrusions to create the crankshaft counterweights and the complex internal geometry of the piston skirt. Parametric design ensured that any dimensional tweaks updated dynamically across the component.



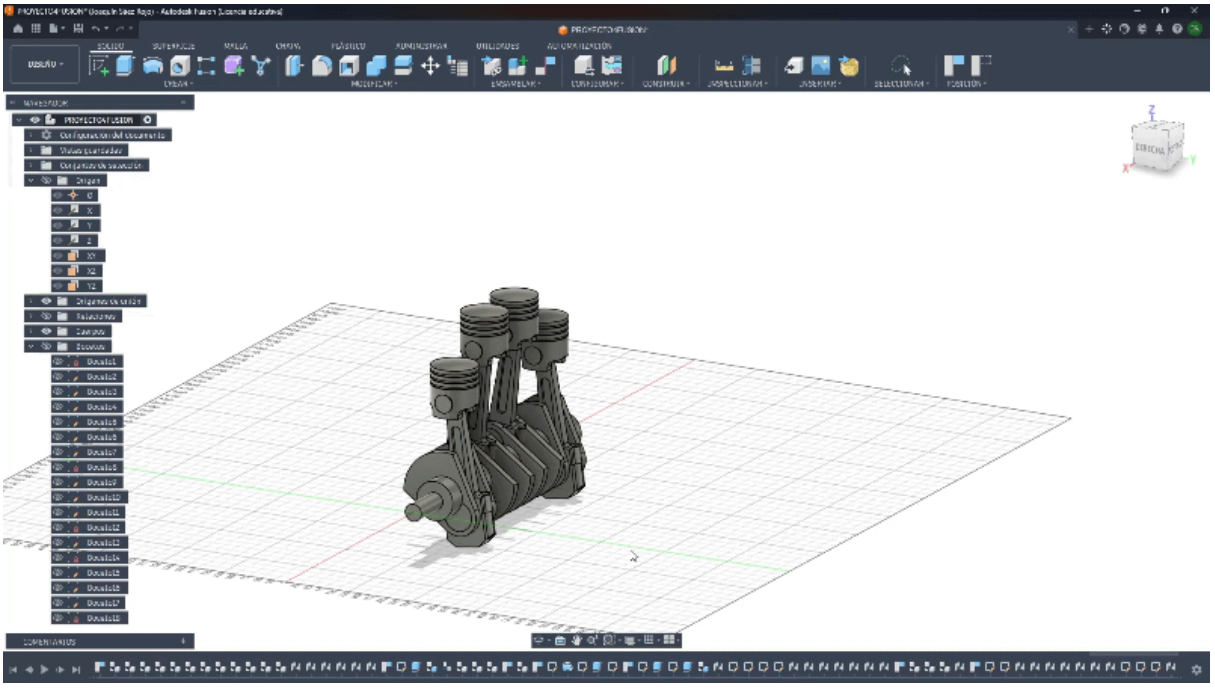




2. Kinematic Assembly and Motion Validation

Once the parts were modeled, the core challenge was assembling them to replicate real-world engine timing and movement without mechanical interference.

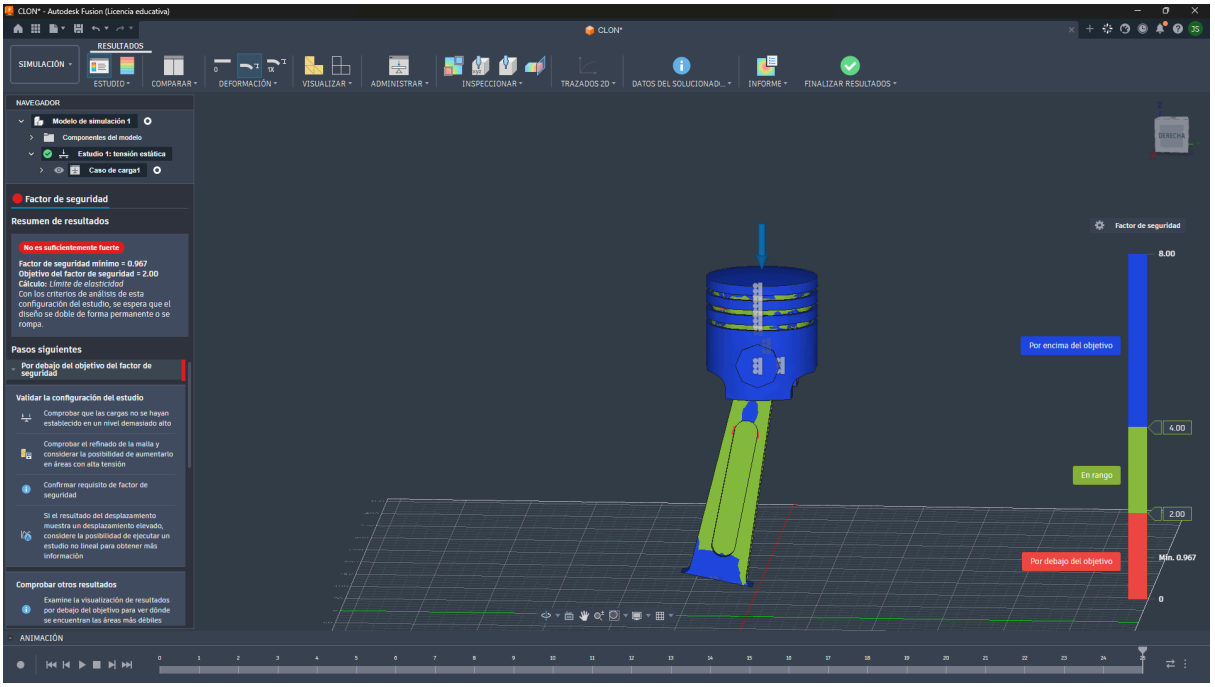
- **Solution:** I established a hierarchy of mechanical joints. I applied a **Revolute Joint** to the crankshaft to allow continuous rotation, **Cylindrical Joints** for the piston pins, and **Slider Joints** to constrain the pistons strictly to the vertical Z-axis (simulating the cylinder walls).



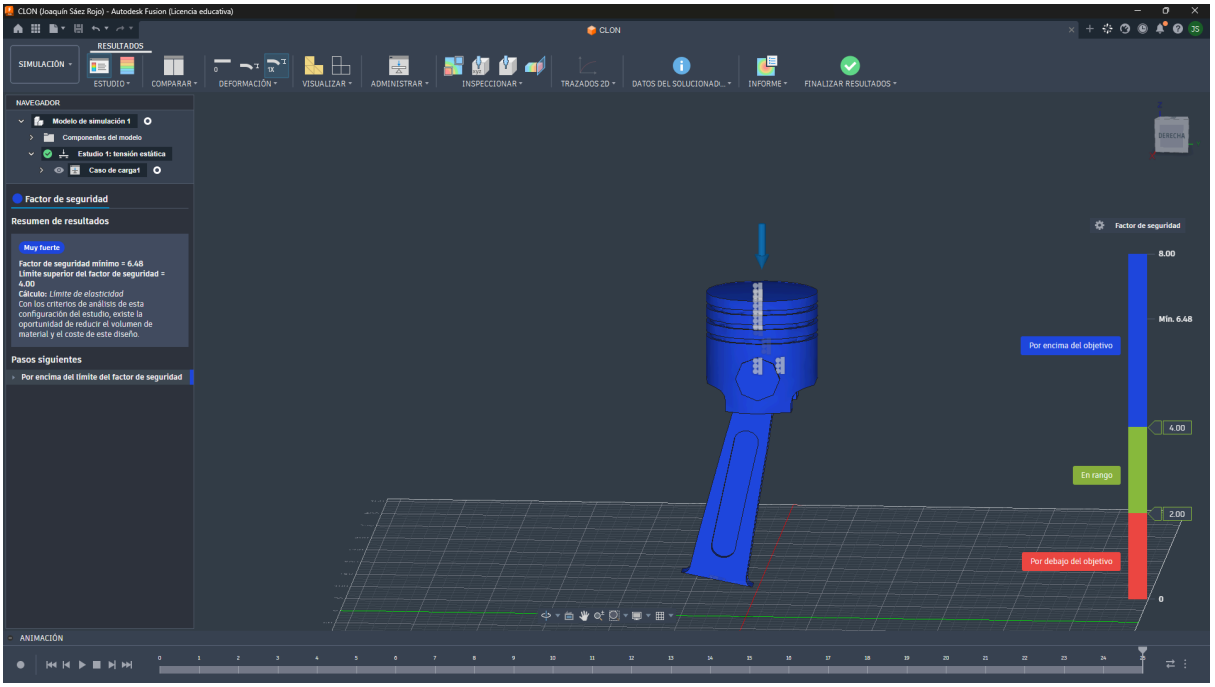
3. Component Stress Analysis (FEA - Bonus Validation)

As a final validation phase, I isolated the piston and connecting rod assembly to test its structural integrity during the exact moment of combustion (Top Dead Center).

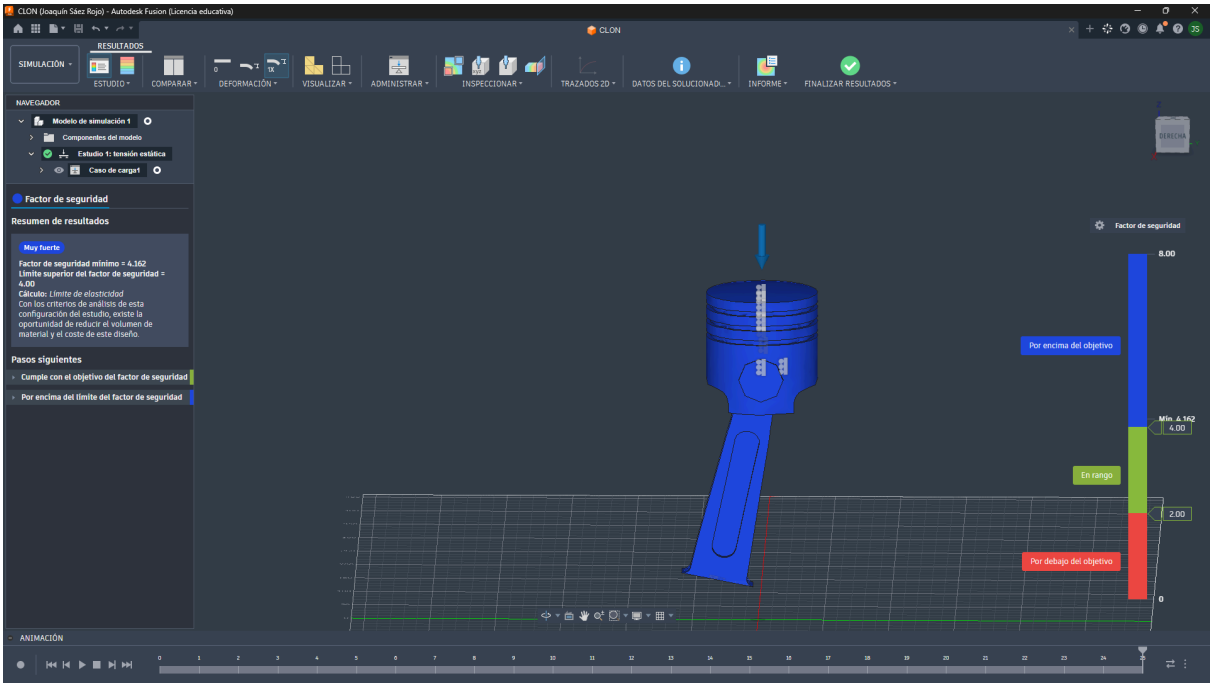
- **Setup:** I applied a 35,000 N downward force on the piston crown and ran a material trade-off study comparing standard Steel (which resulted in structural failure) against **AISI 4142 Alloy Steel** and Titanium-6Al-4V alloy, which both successfully absorbed the load and kept the Safety Factor within a viable range for mass production.



(Standard Steel)



(AISI 4142 Alloy Steel)



(Ti-6Al-4V Alloy)

TECHNICAL CHALLENGES & SOLUTIONS

- **Dimensional Discrepancies and Reverse Engineering:**
  Several reference blueprints contained omitted or slightly inaccurate dimensions. To ensure maximum geometric precision, I extracted the 2D technical views and conducted a manual dimensional correlation study on paper. By analyzing geometric proportions and structural relationships, I successfully deduced the exact missing measurements (such as the specific length of the connecting rod's I-beam recess) prior to 3D modeling.
- **Kinematic Over-constraining and DOF Mapping:**
  During the initial assembly phase, the mechanism experienced kinematic locking because rigid connections were inadvertently applied to rotational nodes. To resolve this, I hand-drew a mechanical linkage diagram of the piston assembly to map out the exact Degrees of Freedom (DOF) required at each connection point. Based on this analysis, I reconfigured the joint hierarchy—properly assigning revolute, cylindrical, and slider joints—which eliminated the binding and achieved the flawless mechanical motion demonstrated above.

- **Isolating Geometry for FEA:**
  Simulating the entire engine assembly at once would require massive computational power. I solved this by deriving only the critical sub-assembly (Piston + Rod) into the Simulation workspace. To account for the missing engine block, I applied a **Frictionless Constraint** to the piston walls, forcing realistic vertical compression and preventing artificial sideways bending under the 35,000 N load.

DEMONSTRATION AND RESULTS

The project resulted in a fully functional, mathematically accurate digital prototype of a 2.0L engine. The kinematic assembly proves the mechanical viability of the design, while the FEA validation confirms that the chosen steel alloys can withstand standard automotive combustion cycles.

Future Roadmap:

- Modeling the engine block, cylinder head, and dual overhead camshafts (DOHC) to complete the powertrain.
- Running an Event Simulation (Dynamic FEA) to analyze stresses while the crankshaft is actively rotating.